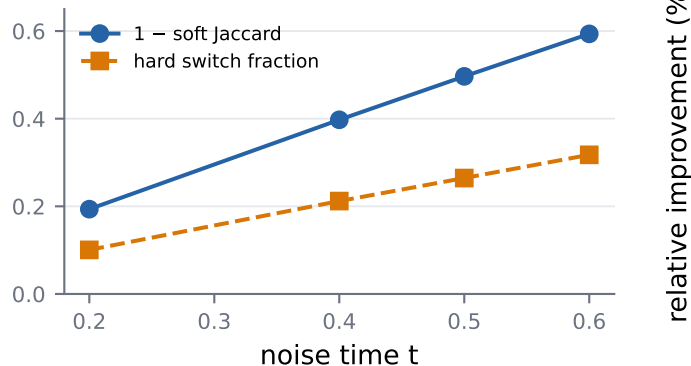


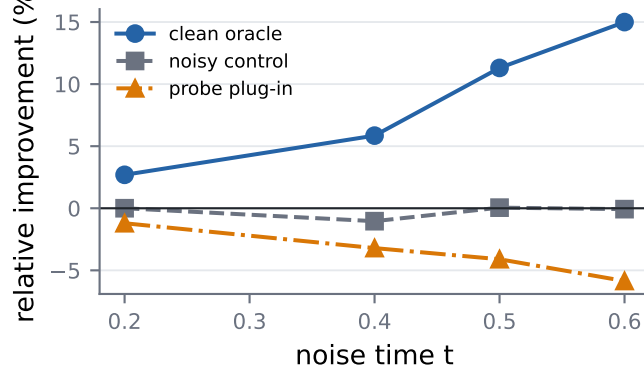
# All-pair clean-topology attribution

Frozen dynamic checkpoint; complete directed non-self pair support; zero optimizer steps

## A Clean/noisy topology disagreement



## B Residual-energy improvement



## C Frozen-probe predictability

